

Cupcast II-V: Held-Out Validation and a Comparison Against the Cupcast I Baseline on the 2026 World Cup

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Abstract

We evaluate the dynamic state-space model of Paper II against the static Cupcast I baseline on the matches of the 2026 World Cup that have actually been played, both models trained strictly before kickoff. On the 60 completed group matches the new model improves out-of-sample log-loss from 0.869 to 0.856 and is better on Brier score and the ranked probability score as well. A pre/post ablation shows the bounded context layer (travel, host, altitude) improves all three scores further; an honest counter-result is that cross-validated hyperparameter tuning, while it lowers the internationals cross-validation loss, *degrades* World Cup performance—the model’s default priors generalise better than tuned ones. We close with the headline forecast and a sharpened version conditioned on the realised knockout bracket.

1 Protocol

Both models are fitted to international results before the tournament cutoff of 11 June 2026; the 60 subsequently completed group matches are a clean held-out test set. We report three strictly proper scoring rules [2] over the three-way (home/draw/away) outcome: logarithmic loss, the multiclass Brier score, and the ranked probability score (RPS). Lower is better; the uniform forecast $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ scores log-loss $\ln 3 = 1.099$. The Cupcast I baseline is its frozen pipeline, queried through a name-mapping layer; the comparison is restricted to matches both models can score.

2 The rebuild improves the metric

Table 1 reports the held-out scores. The dynamic model beats the static baseline on every metric, and both beat the uniform forecast decisively.

Forecaster	Log-loss	Brier	RPS
Cupcast II (dynamic + clubform)	0.856	0.515	0.157
Cupcast I (static baseline)	0.869	0.523	0.160
Uniform	1.099	0.667	0.233

Table 1: Held-out scoring on the 60 played 2026 World Cup group matches. Lower is better.

The improvement is modest ($\sim 1.4\%$ in log-loss) but consistent across all three rules. It is corroborated out of sample on 772 historical international matches under rolling-origin cross-validation, where the dynamic model scores log-loss 0.910 (RPS 0.175) against 1.099 for uniform, on par with a static weighted Dixon–Coles baseline. The most visible behavioural

change is that the Cupcast I overrating of recent competitive form is corrected: the title probability the static model assigned to its favourite falls from an inflated 16.3% to a market-plausible 10.2%.

3 The context layer earns its place

We ablate the bounded context covariates of Paper II by scoring the same held-out matches with and without them (host advantage routed entirely through the context layer so the comparison is clean). Table 2 shows that travel, host, and altitude together improve all three scores.

	Log-loss	Brier	RPS
Model only	0.865	0.521	0.160
Model + context	0.860	0.518	0.158

Table 2: Pre/post-context ablation on the 60 played group matches.

The signals are concrete: Mexico plays all three group games at home at altitude (Mexico City $\approx$ 2240 m, Guadalajara $\approx$ 1560 m), and Iran faces a 1,553 km travel leg between matches. Each effect is hard-capped and provenance-logged, so the gain is earned without letting any single signal dominate.

4 An honest counter-result: tuning overfits the wrong regime

We tuned the model’s prior hyperparameters by Optuna [1] under rolling-origin cross-validation on past internationals. The search found parameters with a *better* cross-validation loss (0.842 versus the defaults), but those parameters *worsened* the World Cup hold-out (log-loss 0.865, up from the untuned 0.856). The internationals folds—qualifiers and friendlies—are a different regime from the tournament (neutral venues, fresh squads, knockout intensity), and tuning to them does not transfer. The practical conclusion is to keep the sensible default priors, which are robust and remain the best configuration on the actual test.

5 The forecast

The production configuration (default priors with the model’s host-advantage term) yields the championship probabilities of Table 3 from 50,000 simulated tournaments. The bounded context covariates are applied and validated at the level of individual match predictions (Section 3); the tournament Monte Carlo itself carries host advantage through the model’s γ term, so the headline probabilities do not depend on path-specific travel that a simulated bracket cannot yet resolve.

The penalty-specific knockout shootout edge of Paper II is included here; its effect is small by construction—shootouts decide only a fraction of knockout paths—nudging penalty-strong sides such as Colombia and France up by a few tenths of a percentage point while leaving the ordering of the favourites unchanged. That the headline is robust to it is reassuring, not disappointing: a model whose forecast swung on coin-flip shootouts would be the more suspect one.

Team	Champion	Team	Champion
Brazil	20.1%	Netherlands	5.5%
Spain	13.2%	Belgium	4.6%
England	11.9%	Colombia	4.5%
Argentina	10.3%	Germany	2.6%
Portugal	8.4%	Switzerland	1.7%
France	6.0%	Uruguay	1.5%

Table 3: Cupcast II championship probabilities for the 2026 World Cup (top teams), from 50,000 Monte Carlo tournaments with the held-out model.

6 Knockout forecast on the actual draw

With the group stage complete, the forecast of Table 3 can be sharpened: rather than simulating the groups, we seed 50,000 tournaments directly from the *actual* Round-of-32 bracket and play forward. The bracket is reconstructed from the official final standings and validated slot-by-slot against the real draw—a check that exposed our Annex-C best-thirds allocator as a heuristic which mis-pairs three of the sixteen third-placed slots, so the third-place pairings are taken from the real draw rather than the allocator. The held-out model and the penalty shootout edge are otherwise unchanged.

Team	Champion	Team	Champion
Brazil	19.4%	Netherlands	6.1%
Argentina	15.2%	Colombia	5.4%
England	13.0%	Belgium	4.1%
Spain	12.1%	Germany	2.6%
Portugal	6.8%	Switzerland	1.6%
France	6.2%	Morocco	1.3%

Table 4: Championship probabilities conditioned on the actual Round-of-32 draw, from 50,000 Monte Carlo tournaments seeded with the real bracket (validated against the official draw).

Conditioning on the real draw reshapes the title race. Argentina rises from 10.3% to 15.2%: it landed in the softest quadrant—Cape Verde in the Round of 32—and now carries the field’s highest semifinal probability (49.8%). England firms to 13.0% on a favourable path, while Brazil remains the nominal favourite (19.4%) with Argentina close behind. Uruguay, eliminated in the group stage, leaves the title picture entirely.

7 Discussion

Two of the redesign’s premises are validated on the real tournament: a lineup-aware, dynamically-evolving model beats the static results-only baseline, and bounded contextual signals add genuine predictive value. A third, more sobering, lesson is that aggressive hyperparameter tuning can overfit a proxy distribution; the disciplined defaults win. Remaining work—weather and injury covariates, and a learned (rather than shadow) morale signal subject to the same ablation gate—can only be admitted to the production forecast if it clears the same held-out bar these results were held to.

References

- [1] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2019.
- [2] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.